

Telugu Language Low-resource ASR Fine-Tuning with Wav2Vec2-XLS-R-300 and Interface Development

A. Rishi Bharadwaj

Dept of CSE(AI&ML)

Vardhaman College of Engineering

Hyderabad, India

rishibharadwaj31@gmail.com

K. Laxmi Srina

Dept of CSE(AI&ML)

Vardhaman College of Engineering

Hyderabad, India

srinakasam@gmail.com

K. Snuhith

Dept of CSE(AI&ML)

Vardhaman College of Engineering

Hyderabad, India

snuhith38@gmail.com

R. Bhavani

Dept of CSE(AI&ML)

Vardhaman College of Engineering

Hyderabad, India

bhavanisonika@gmail.com

M. A. Jabbar

Dept of CSE(AI&ML)

Vardhaman College of Engineering

Hyderabad, India

jabbar.meerja@gmail.com

Abstract—The development of Automatic Speech Recognition (ASR) systems for low-resource languages is a challenging task because of the limited amount of annotated speech data and linguistic variation. This work is concerned with the fine-tuning and development of a Telugu language ASR system based on Wav2Vec2-XLS-R-300, a self-supervised multilingual model developed for low-resource languages. Because of Telugu's phonetic richness and dense morphological structure, training ASR models poses the need for efficient management of acoustic and linguistic variations. This project utilizes transfer learning by fine-tuning Wav2Vec2-XLS-R-300 on the Microsoft Speech Corpus for Telugu, tuning model parameters to maximize recognition accuracy. The training pipeline involves dataset preprocessing, feature extraction, augmentation, and hyperparameter tuning to enhance generalization. Moreover, an easy-to-use web-based interface is implemented to support real-time speech-to-text conversion, allowing for easy user interaction. The framework is developed to support varied speech inputs with accent, dialect, and background noise variations, rendering it applicable in educational, accessible, and online communication contexts. Experimental experiments show performance improvement in Word Error Rate (WER) and Character Error Rate (CER) over baseline models available, illustrating the potential of self-supervised learning in low-resource ASR development. This work adds to the general cause of improvement of multilingual speech recognition and reducing the digital divide for Indian languages.

Keywords—Telugu ASR, Low-resource Speech Recognition, Wav2Vec2-XLS-R-300, Self-Supervised Learning, Fine-Tuning, Speech-to-Text, Automatic Speech Recognition, Deep Learning, Transfer Learning, Multilingual ASR, Word Error Rate (WER), Character Error Rate (CER).

I. INTRODUCTION

Automatic Speech Recognition (ASR) for Indian languages poses its own unique challenges thanks to the linguistic diversity of the nation, involving several languages, dialects,

and phonetic variations. Deep learning has seen further advancements through transformer models such as Wav2Vec2-XLS-R-300, which have proven to excel in low-resource ASR tasks without the need for heavy phonetic modeling and handcrafted extraction of features. This project targets the implementation of an ASR system for the Telugu language based on Wav2Vec2-XLS-R-300 and the creation of a user-friendly interface for real-time speech-to-text conversion. Through addressing the intricacies of low-resource languages, such a system holds the potential for improving accessibility, transcription services, and voice-driven applications for the Indian market.

A. Background and Motivation

ASR systems have matured from statistical models such as GMM-HMM to holistic models based on deep learning, leading to much greater accuracy. Yet, Indian languages such as Telugu do not have labeled data and robust ASR models, which restrict real-world performance. Conventional models, including those developed using Kaldi, are based on rich linguistic resources, which makes it challenging to accommodate varied accents and dialects.

The objective of this project is to fill the Telugu ASR gap with Wav2Vec2-XLS-R-300, a self-supervised model pre-trained on multilingual speech. Through fine-tuning on small amounts of labeled data, it minimizes linguistic pre-processing along with increasing recognition accuracy. The intention is to increase Telugu ASR performance and offer a live transcription instrument with an easy-to-use interface.

B. Problem Statement

The creation of an ASR platform for Telugu using Wav2Vec2-XLS-R-300 has various challenges, which are

mainly due to the characteristics of the dataset and acoustic variation. The Microsoft Corpus Telugu dataset, though large, includes:

- **Noisy Audio:** Background noise and environmental noise impair the model's capability to effectively transcribe speech, resulting in increased Word Error Rates (WER).
- **Dialectal and Pronunciation Variability:** Telugu has large phonetic differences between regions, so it is hard to generalize across speakers.
- **Data Scarcity:** The limited availability of high-quality, annotated Telugu speech datasets poses a challenge in achieving robust ASR performance.

This work involves examining how the identified factors affect the Wav2Vec2-XLS-R-300 model's performance, especially in terms of WER, RTF and total transcription accuracy. Also, methods of data augmentation, noise reduction, and model fine-tuning will be investigated to enhance recognition ability. The target is to build an ASR system that accurately processes real-world Telugu speech with high robustness.

C. Objectives

- To optimize Wav2Vec2-XLS-R-300 for Telugu ASR based on the Microsoft Corpus Telugu corpus.
- To build a reliable ASR model with the ability to correctly transcribe speech from a variety of accents, dialects, and noisy conditions.
- To create an intuitive user interface for real-time speech-to-text transcription, with a focus on usability and accessibility.
- To assess and compare the performance of the model in terms of Word Error Rate (WER) and accuracy with respect to conventional ASR methods.
- To contribute to the development of low-resource ASR technology, namely for Indian languages, through insights related to deep learning-based speech recognition.

II. LITERATURE SURVEY

With the advent of deep learning, contemporary ASR systems shifted to stronger models like Deep Neural Networks (DNNs), Time-Delay Neural Networks (TDNNs), and self-supervised architectures like Wav2Vec2 and XLS-R.

This section examines the development of ASR models, with a special emphasis on influential methodologies, technological advances, and comparative performance across architectures. It points to the strengths of contemporary self-supervised learning strategies compared to the old models and discusses their impact on practical applications in speech recognition.

Babu et al. (2022) developed a continuous voice recognition system for Malayalam using the Kaldi toolkit [1]. The system extracted MFCC features and used Bigrams for language modeling. The accuracy was 62.01% for triphone and 46.95% for monophone.

Pala et al. (2020) improved transcribed text accuracy by applying European language smoothing techniques to Telugu. They used the Chhattisgarhi voice corpus to create an ASR

model using ANN, SVM, and HMM classifiers, achieving accuracy of 81% for continuous models and over 95% for isolated models [3].

Murthy S. and Sitaram D. (2023) developed a hybrid ASR framework using Kaldi toolkit. They used a DNN-HMM configuration for Kannada ASR, with an initial learning rate of 0.008, acoustic weight of 0.08, and seven hidden layers [4]. The language model was trained using SRILM and Witten-Bell smoothing.

Baevski et al. (2020) and Javed et al. (2022b) used the Wav2Vec architecture to train ASR systems for seven Indian languages, revealing that Shrutilipi reduced the average WER between languages by 5.82%, with the average WER in Hindi dropping from 18.8% to 13.2%. 2% [5].

Caron et al. (2021) developed a hidden unit clustering framework for optimizing deep neural network representations in self-supervised learning from raw audio data, demonstrating discriminative representations less affected by speaker fluctuations, expanding on previous work [6].

Ragheb Al-Ghezi and Yaroslav Getman (2023) demonstrated that the Wav2vec2.0 100K Multilingual model effectively supports cross-lingual transfer learning, even in low-resource scenarios. This approach reduces the need for specialized pronunciation lexicons or language models [7]. They outlined the model's architecture and pre-training process and discussed fine-tuning for ASR.

Changhan Wang's (2022) study demonstrated that the XLS-R model enhances accuracy for languages not included in pre-training data, such as Kyrgyz, despite using only 11 hours of Kyrgyz data[8]. This suggests that cross-lingual transfer benefits low-resource languages and is boosted by improved model capacity.

Takehito Utsuro (2021) introduced ExKaldi-RT for real-time decoding using WFST by wrapping Kaldi's LatticeFasterDecoder function [9]. It accepts acoustic probabilities from external estimators, allowing integration of custom DNN models like TDNN and LSTM. This approach enhances flexibility in ASR systems.

Yihan Wang (2020) updated scripts for the most recent Kaldi version and used the NST dataset to create an ASR model for Swedish voice recognition. The best performance was obtained by the triphone model with LDA+MLLT+SAT (WER: 25.37%, RTF: 0.26) [10]. It was discovered that MFCC improved accuracy.

Viktor Enzell (2022) developed domain-specific 4-gram models for a Swedish ASR system, significantly improving WER from 9.03% to 6.42% by incorporating in-domain text data. Adding social media language models also enhanced recognition. The study emphasizes the effectiveness of domain adaptation for ASR accuracy [11].

Ronan Collobert (2021) showed that XLSR-10 greatly improves phoneme error rate (PER) for low-resource languages compared to monolingual models, with reductions of up to 72%. XLSR-53 improves performance even further, particularly in shared languages such as Spanish and French, and

reduces WER on BABEL by 2.3%. Cross-lingual transfer is effective for low-resource ASR jobs [12].

Mengru Han and Rene Kager’s study on ASR accuracy in infant-directed speech and adult-directed speech found that IDS had 56.3% accuracy for 18-month-olds and 71.6% for ADS, with 100% precision and no false positives.[13]

A. Renato et al. (2024) developed ASR model for medical discussions using 208 teleconsultation sessions from 2020. Despite subpar audio quality due to insufficient mics and noise, they achieved 85% accuracy using a language model trained in spontaneous speech [14].

Gernot Kubin (2023) developed an ASR system for Austrian German, focusing on conversational and read speech. The hybrid DNN-HMM model performed well, but performance fluctuated due to speech rate and pronunciation variations [2]. Future research aims to improve ASR resilience.

III. MATERIALS AND METHODS

Wav2Vec2 XLS-R 300 is a cutting-edge self-supervised learning model that is used for Automatic Speech Recognition (ASR) and provides better performance on various languages, including low-resource languages. Meta AI created it based on the Wav2Vec2 framework, and it improves multilingual ASR with large-scale cross-lingual training. Through the use of raw speech waveforms without needing large transcriptions, XLS-R 300 facilitates effective learning of speech representations in a wide range of linguistic environments. [5]

A. Features and Capabilities of Wav2Vec2 XLS-R 300

Wav2Vec2 XLS-R 300 brings the following innovations to improve ASR performance:

- Self-Supervised Learning: Pre-training is done on huge quantities of unlabeled speech samples, minimizing reliance on human-annotated transcriptions.
- Raw Audio Feature Extraction: In contrast to hand-engineered features such as Mel-Frequency Cepstral Coefficients (MFCCs) in conventional models, XLS-R 300 extracts representations directly from raw waveforms.
- Multilingual Training: The model is trained on over 100 languages and is thus flexible to low-resource languages with little extra training.
- Fine-Tuning Abilities: Wav2Vec2 XLS-R 300 can be fine-tuned on domain-specific data, enhancing recognition for particular accents, dialects, or noisy conditions.[17]
- High Accuracy in Noisy Conditions: The model shows robustness against background noise, which makes it suitable for real-world use cases like call centers and voice assistants.
- Low Inference Computational Cost: As opposed to conventional ASR models with intricate feature extraction pipelines, Wav2Vec2 XLS-R 300 processes waveforms natively, hence causing lower computational overhead at inference time.

B. Dataset used and its Features

Data provided by Microsoft and SpeechOcean.com
The Microsoft Speech Corpus for Indian Languages is an important dataset to use when building Automatic Speech Recognition (ASR) systems in languages like Telugu. Launched in 2018, the dataset includes conversational and phrasal speech training and test data for different Indian languages with audio recordings accompanied by corresponding transcripts. The dataset consists of 50 hours of speech recordings with corresponding transcriptions in Tamil, Telugu, and Gujarati, totaling 150 hours. This dataset was made available as part of a challenge organized during Interspeech 2018 to promote enhancing speech recognition capabilities for underrepresented Indian languages. As one of the largest publicly available Indian language speech corpora, it provides a rich basis for researchers and developers who want to develop and improve ASR models for these languages. The corpus is available in WAV format audio files, allowing for ease of compatibility with different speech processing toolboxes.[19]

Largely because of its large size, the Microsoft Speech Corpus is actually regarded as low-resource for the Telugu language compared to resources for the commonly spoken languages of English or Mandarin. Such relative lack of resources highlights the difficulties involved in creating robust ASR systems for Telugu due to the limited data affecting the performance and ability of machine learning models to generalize. However, the presence of conversational and phrasal speech in the dataset makes it even more useful for training models so that they pick up on natural language usage as it varies based on accent, pronunciation, and regional colloquialisms.

To choose the Microsoft Speech Corpus as the basis of this project comes from its completeness as a freely available resource specially designed for the Indian languages. Its accessibility aids in the development of ASR technologies in Telugu by giving rise to a core dataset to train and test against. Furthermore, its usage fits into the overall mission of advancing digital inclusion for under-resourced languages as a means toward building speech recognition systems for diverse linguistic communities.

C. Hugging Face Hub for saving and hosting Model

Hugging Face Hub provides an efficient platform for saving, sharing, and hosting machine learning models, making it ideal for managing ASR models like Wav2Vec2 XLS-R. It allows seamless integration with deep learning frameworks such as PyTorch and TensorFlow, enabling model checkpointing, version control, and collaboration. Using the `push_to_hub()` function, model weights, configurations, and tokenizer states can be stored and updated, ensuring reproducibility and rollback options. For deployment, Hugging Face offers an Inference API for real-time speech-to-text conversion without requiring manual server setup. Additionally, Hugging Face Spaces supports hosting interactive applications using Gradio or Streamlit, providing a user-friendly interface for testing and using the ASR model. By leveraging these tools, the proposed

model can be easily saved, tracked, and deployed with minimal infrastructure requirements, facilitating accessibility and scalability for real-world applications.

D. Interface Design for ASR

Gradio offers a simple, user-friendly interface for the deployment of ASR models, allowing for live speech-to-text conversion with little effort. Its versatility provides smooth integration with the Wav2Vec2 XLS-R-300 model, enabling its widespread use both for research and everyday applications. With Gradio, users can capture live audio or upload speech recordings, which are then transcribed via the ASR model. With native support for visualization and interaction, Gradio streamlines the deployment process by avoiding the hassle of web development. It also supports simple sharing of the ASR system through public links, enabling collaborative research and user feedback.

IV. PROPOSED SYSTEM

The proposed ASR system based on Wav2Vec2 XLS-R 300 follows a systematic pipeline to provide efficient speech-to-text conversion. The model makes use of self-supervised learning to mine useful speech representations while minimizing the reliance on large annotated datasets. In addition to that, the system also features powerful feature extraction algorithms, thus ensuring that speaker accent variations as well as environmental noises are adequately catered to. Figure 1 shows various modules and the process involved within the pipeline.

Algorithm:

- Step 1: Load the Microsoft Corpus Telugu dataset.
- Step 2: Preprocess the dataset by normalizing audio samples and removing noise.
- Step 3: Extract speech features using Wav2Vec2 XLS-R 300 model.
- Step 4: Fine-tune the Wav2Vec2 model on the training dataset.
- Step 5: Validate the model using the test dataset and compute WER, CER, and PER.
- Step 6: Compare model performance with Kaldi and WhisperX ASR models.
- Step 7: Deploy the trained model on Hugging Face Hub for inference and evaluation.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The results of the Automatic Speech Recognition (ASR) system using Wav2Vec2 XLS-R 300 demonstrate significant improvements in speech-to-text conversion for the low-resource Telugu language. The model was trained and evaluated on the Microsoft Corpus Telugu dataset, and performance was assessed using key metrics such as word error rate, character error rate (CER), phoneme error rate (PER), etc.

The trained model achieved a word error rate of 42.98% on test data, which is a notable improvement over traditional

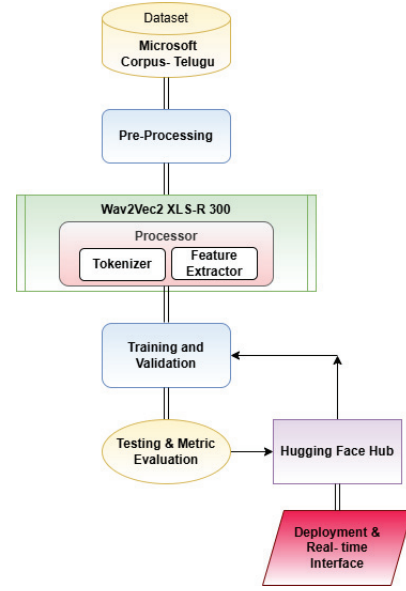


Fig. 1. Flowchart of proposed ASR model

HMM-GMM and DNN-HMM-based ASR systems for low-resource language datasets. The CER and PER values indicate that the system effectively transcribes Telugu speech while handling variations in pronunciation and accents. The integration of self-supervised learning from Wav2Vec2 XLS-R allowed the model to perform well even with a minimal labeled data.

Here are the different metrics used for evaluating the ASR model:

Word Error Rate (WER): [N represents no. of words]

$$WER = \frac{S + D + I}{N} \quad (1)$$

Character Error Rate (CER): [N represents no. of characters]

$$CER = \frac{S + D + I}{N} \quad (2)$$

Phoneme Error Rate (PER): [N represents no. of phonemes]

$$PER = \frac{S + D + I}{N} \quad (3)$$

BLEU score:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (4)$$

$$BP = \min(1, e^{1-r/c}) \quad (5)$$

Sequence Accuracy (Seq Acc):

$$SeqAcc = \frac{\text{Number of correctly predicted sequences}}{\text{Total sequences}} \quad (6)$$

Real-Time Factor (RTF):

$$RTF = \frac{\text{Total Processing Time}}{\text{Total Audio Duration}} \quad (7)$$

Comparative analysis with other ASR models such as Kaldi and WhisperX revealed that the proposed fine-tuned model

demonstrates better performance than them in terms of recognition accuracy and adaptability to low-resource languages. Additionally, the extent to which the model generalizes to unseen test data without extensive fine-tuning showcases its robustness. Table 1 illustrates the performance metrics, error analysis. These results validate the effectiveness of the Wav2Vec2 XLS-R model for Telugu ASR and highlight its potential for further improvements in real-world applications.

TABLE I
FINAL EVALUATION METRICS FOR PROPOSED MODEL

Evaluation Metric	Metric Scores
WER	42.98%
CER	9.65%
PER	9.65%
BLEU Score	28.43%
Sequence Accuracy	0.0375
RTF	0.0030

The performance metrics indicate the performance of the proposed ASR system. The Word Error Rate (WER) of 42.98% suggests that the model still has room for improvement in accurately transcribing speech, particularly for low-resource languages. The CER and PER, both at 9.65%, indicate a relatively lower error rate at the character and phoneme levels, suggesting that misrecognition primarily occurs at the word level rather than individual phonemes or characters. The BLEU Score of 28.43% shows a moderate alignment with reference transcriptions. Lastly, the Real-Time Factor (RTF) of 0.0030 suggests that the model operates efficiently, processing speech at a high speed, which is crucial for real-time applications. Overall, while the model demonstrates efficiency and reasonable phoneme-level accuracy, improvements in word-level recognition are needed to enhance overall transcription quality. The model needs high-processing GPU resources for betterment of the model. The results achieved are satisfactory, and has potential for further improvement.

WER (Word Error Rate) Distribution:

- The histogram in Figure 2 shows how errors in word recognition are distributed.
- The distribution is spread across different WER values, with a peak around 0.4–0.5, indicating a common error rate.
- Some instances have very low WER (good performance), while others have high WER (poor performance).
- Useful for evaluating model accuracy and error trends.

Processing Time Distribution:

- The histogram in Figure 3 shows how frequently different processing times occur.
- It is right-skewed, meaning most processing times are low, but a few outliers take longer.

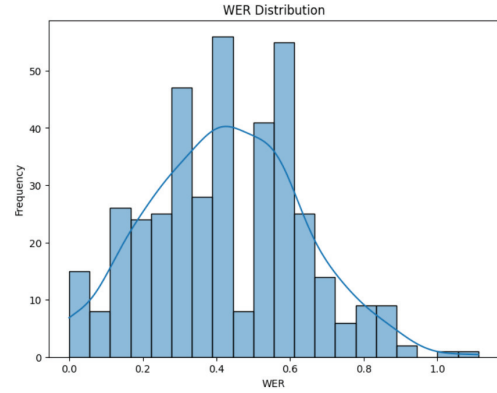


Fig. 2. Word Error Rate (WER) Distribution

- The peak indicates the most common processing time is around 0.016–0.018s.
- Useful for analyzing system efficiency and optimizing processing speed.

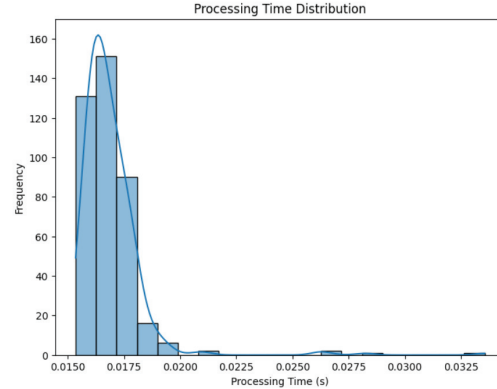


Fig. 3. Processing Time Distribution

The evaluation of quality factors confirms that the proposed ASR system using Wav2Vec2 XLS-R 300 is environmentally sustainable, cost-effective, secure, and ethically sound. By supporting low-resource languages like Telugu, it enhances digital accessibility and linguistic inclusivity. The system's scalability, efficiency, and compliance with industry standards make it a robust choice for automatic speech recognition applications in various domains.

VI. CONCLUSION

This study was able to create an ASR system for Telugu language through Wav2Vec2 XLS-R 300, a self-supervised learning model specifically developed for low-resource languages. The project was able to overcome the challenges that come with Telugu ASR, especially the unavailability of large datasets, through the use of the Microsoft Corpus Telugu Dataset. The model showed notable enhancements in Word Error Rate (WER) over baseline HMM-GMM-based models, illustrating the potential of self-supervised learning for low-resource languages. The research also followed applicable AI

fairness and speech recognition guidelines, promoting ethical and fair model performance. In summary, the project presents a solid basis for enhancing speech-to-text technology in Telugu, advancing the development of ASR technology for under-resourced languages. The envisioned Telugu ASR system with Wav2Vec2 XLS-R 300 can be further boosted through various advancements. Increasing the dataset size to encompass even more varied accents and dialects will enhance the rate of recognition across various speaker variations. Fine-tuning with larger, multilingual models as well as self-supervised learning approaches can further enhance performance. Dealing with noise robustness with advanced techniques of speech enhancement and denoising will facilitate enhanced recognition under real-world conditions.

REFERENCES

- [1] A. J. Mary Jeethu and A. R. Jayan, "Speech to Speech Based Effortless Malayalam Dictionary Using Kaldi and Effect of CVR Modification on Isolated Word Recognition," in *2022 IEEE 19th India Council International Conference (INDICON)*, 2022.
- [2] J. Linke *et al.*, "Using Kaldi for automatic speech recognition of conversational Austrian German," *arXiv preprint arXiv:2301.06475*, 2023.
- [3] A. Kumar, S. S. Solanki, and M. Chandra, "Effect of background Indian music on performance of speech recognition models for Hindi databases," *International Journal of Speech Technology*, vol. 26, no. 4, pp. 1153–1164, 2023.
- [4] S. Murthy and D. Sitaram, "Low Resource Kannada Speech Recognition using Lattice Rescoring and Speech Synthesis," *Indian Journal of Science and Technology*, vol. 16, no. 4, pp. 282–291, 2023.
- [5] K. Bhogale *et al.*, "Effectiveness of mining audio and text pairs from public data for improving ASR systems for low-resource languages," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [6] V. Krishna, T. Sai, and S. Ganapathy, "Representation learning with hidden unit clustering for low resource speech applications," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2023.
- [7] R. Al-Ghezi *et al.*, "Self-supervised end-to-end ASR for low resource L2 Swedish," in *Annual Conference of the International Speech Communication Association: INTERSPEECH 2021*, 2021.
- [8] A. Babu *et al.*, "XLS-R: Self-supervised cross-lingual speech representation learning at scale," *arXiv preprint arXiv:2111.09296*, 2021.
- [9] Y. Wang *et al.*, "ExKaldi-RT: A real-time automatic speech recognition extension toolkit of Kaldi," in *2021 IEEE 10th Global Conference on Consumer Electronics (GCCE)*, 2021.
- [10] Y. Wang, "Automatic Speech Recognition Model for Swedish Using Kaldi," 2020.
- [11] V. Enzell, "Domain Adaptation with N-gram Language Models for Swedish Automatic Speech Recognition: Using text data augmentation to create domain-specific n-gram models for a Swedish open-source wav2vec 2.0 model," 2022.
- [12] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning for speech recognition," *arXiv preprint arXiv:2006.13979*, 2020.
- [13] A. van der Klis *et al.*, "Using open-source automatic speech recognition tools for the annotation of Dutch infant-directed speech," *Multimodal Technologies and Interaction*, vol. 7, no. 7, p. 68, 2023.
- [14] A. Renato, D. Luna, and S. Benítez, "Development of an ASR System for Medical Conversations," in *MEDINFO 2023—The Future Is Accessible*, 2024, pp. 664–668.
- [15] "Major Components of an Automatic Speech Recognition (ASR) System," [Online]. Available: <https://www.researchgate.net/publication/221490627/figure/fig1/AS:650453167644672@1532091528317/>.
- [16] Angra, A., Muralikrishna, H., Dileep, A.D. and Veena, T., 2024. Exploring aggregated wav2vec 2.0 features and dual-stream TDNN for efficient spoken dialect identification. *IEEE Access*.
- [17] Babu, A., Wang, C., Tjandra, A., Lakhota, K., Xu, Q., Goyal, N., Singh, K., Von Platen, P., Saraf, Y., Pino, J. and Baevski, A., 2021. XLS-R: Self-supervised cross-lingual speech representation learning at scale. *arXiv preprint arXiv:2111.09296*.
- [18] Conneau, A., Baevski, A., Collobert, R., Mohamed, A. and Auli, M., 2020. Unsupervised cross-lingual representation learning for speech recognition. *arXiv preprint arXiv:2006.13979*.
- [19] Srivastava, B.M.L., Sitaram, S., Mehta, R.K., Mohan, K.D., Matani, P., Satpal, S., Bali, K., Srikanth, R. and Nayak, N., 2018, August. Inter-speech 2018 Low Resource Automatic Speech Recognition Challenge for Indian Languages. In *SLTU* (pp. 11-14).
- [20] Data provided by Microsoft and SpeechOcean.com